

Thonny - <untitled> @ 45:51

```
2 for i in range(0,10): #range(start,end,increment) - start is inclusive, end is exclusive
3     print("s") #adds a new line after each print
4 #END for
5
6 #Example 2: Print a row of 20 "*" to the screen
7 for i in range(0,20):
8     print("*", end="") #prevent print() function from adding a new line by using end parameter
9 #END for
10
11 #Example 3: Print numbers 0 to 4
12 for i in range(0,5): #0 is inclusive, 5 is exclusive
13     print(i)
14 #END for
15 print("") #add a break line
16
17 #Example #4: Count by 2s from 0-10: 0, 2, 4, 6, 8
18 for i in range(0, 10, 2):
19     print(i)
20 #END for
21 print("") #add a break line
22
23 #Example #5: print the even numbers 0 -> user's choice
24 num=int(input("Please enter an integer:"))
25 for i in range(0, num, 2):
26     print(i)
27 #END for
28 print("") #add a break line
29
30 #Example #6: Calculate the sum of 1->user's number
31 sum=0 #create a variable to hold a growing sum
32 for x in range(1,num):
33     sum=sum+x
34     print("Current sum value is:",sum)
35
36 print("The sum from 1 to ",num,"is: ",sum)
37 print("")
38
39 #Example #7: Use math library to print square root 0-5
40 import math #math library has more than the basic arithmetic operators
41
```

Shell

```
0
2
Current sum value is: 1
Current sum value is: 3
```

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```
18
19 #Example #2: Temperature Monitoring Using Relational Operators as Loop Condition
20 current_temp = 15.0
21 target_temp = 22.0
22
23 #If the line below said "if" instead of "while", the block runs only once if current_temp < target_temp
24 #Because it is a while loop, the block instead runs over and over, as long as the condition is true
25
26 while current_temp < target_temp:
27     print(f"Current: {current_temp} C, heating...") #This line belongs to the loop. Indent
28     current_temp = current_temp + 0.5 # Simulate heating and update condition
29 #END while
30 print("Target temperature reached!") #This line does not belong to the loop. No indentation
31 print("") #adding a break line
32
33 #Example #3: User Input Validation with Logical Operator
34 # This example keeps asking the user for a number until they enter a value 0->200
35
36 entry=int(input("Please enter an integer between 0 and 200"))
37
38 while entry < 0 or entry > 200: #if the entry is invalid (<0 or >200) then we have to ask again
39     print("Invalid number! Please try again.")
40     entry=int(input("Please enter an integer between 0 and 200")) #update condition
41
```

Shell

```
Current: 15.0 C, heating...
Current: 15.5 C, heating...
Current: 16.0 C, heating...
Current: 16.5 C, heating...
Current: 17.0 C, heating...
Current: 17.5 C, heating...
Current: 18.0 C, heating...
Current: 18.5 C, heating...
Current: 19.0 C, heating...
Current: 19.5 C, heating...
Current: 20.0 C, heating...
Current: 20.5 C, heating...
Current: 21.0 C, heating...
Current: 21.5 C, heating...
Target temperature reached!

Please enter an integer between 0 and 20050
Correct entry. Thank you!

Please enter your name:rumi
Hello rumi

>>>
```

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```
Shell
>>> %Run -c $EDITOR_CONTENT
$
$
$
$
$
$
$
$
$
$
*****0
1
2
3
4
0
2
4
6
8
Please enter an integer:4
0
2
Current sum value is: 1
Current sum value is: 3
Current sum value is: 6
The sum from 1 to 4 is: 6
Square root of 0 is: 0.00
Square root of 1 is: 1.00
Square root of 2 is: 1.41
Square root of 3 is: 1.73
Square root of 4 is: 2.00
>>>
```

```
Thonny - <untitled> @ 16:32
Jawad_Py-2_10-6-26_3.py  <untitled> *
Assistant
1  """
2  A condition Contolled While loop that echoes user input until 0 is entered.
3  Keyvani, 10-1-2026
4  """
5
6  # Get the first number before the loop starts
7  num = int(input("Enter a number (0 to stop): "))
8
9  # Keep looping as long as the user hasn't entered 0
10 while num != 0: #We use while loop because the number of iteration is unknown and is condition dependent
11     print(num)
12     num = int(input("Enter a number (0 to stop): ")) #This line updates the loop condition and helps avoid infinite loop
13 #END while
14
15 # Loop has ended - the user entered 0
16 print("Goodbye") #Print goodbye
Shell
>>> %Run -c $EDITOR_CONTENT
Enter a number (0 to stop): 4
4
Enter a number (0 to stop): 2
2
Enter a number (0 to stop): 0
Goodbye
>>>
```